

Python Básico

I. Algoritmo

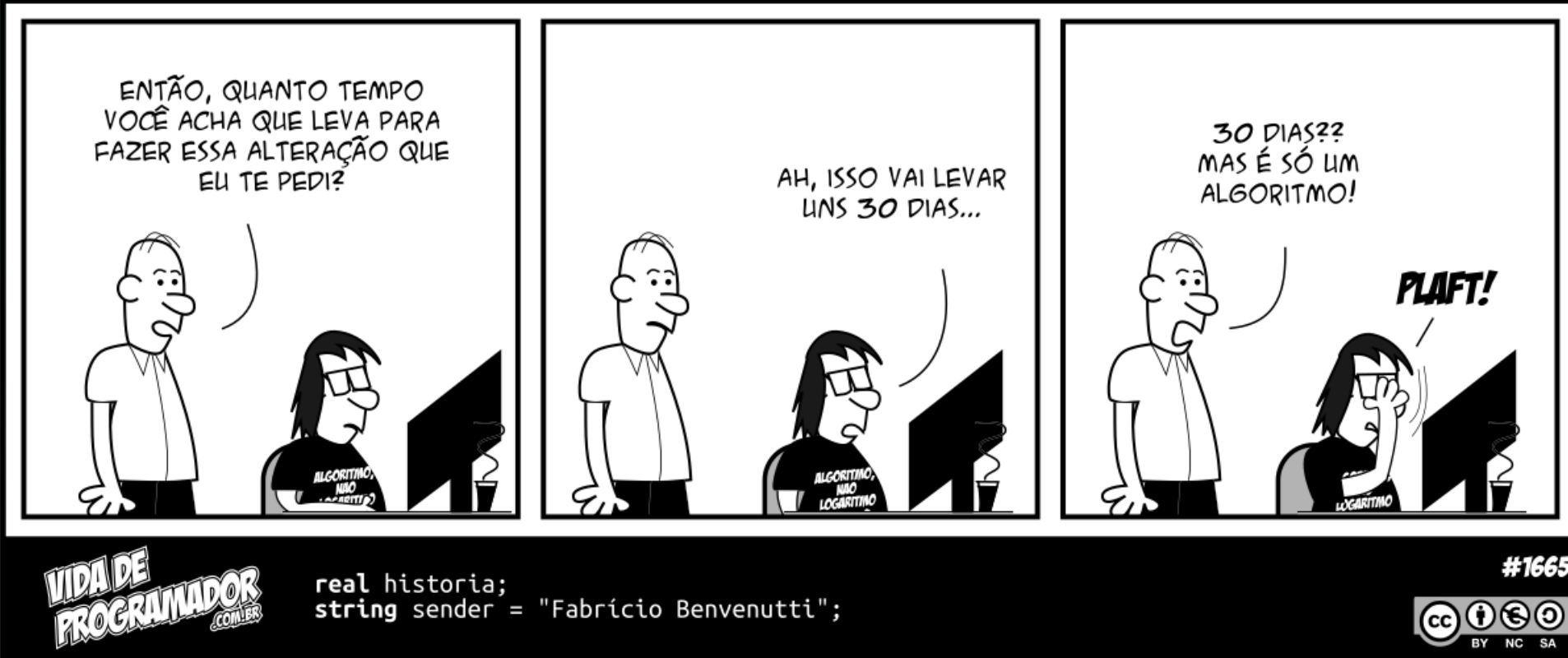
Introdução a algoritmos 1 (conceito e história)

Guilherme Ferreira Silveira

5 de abril de 2026

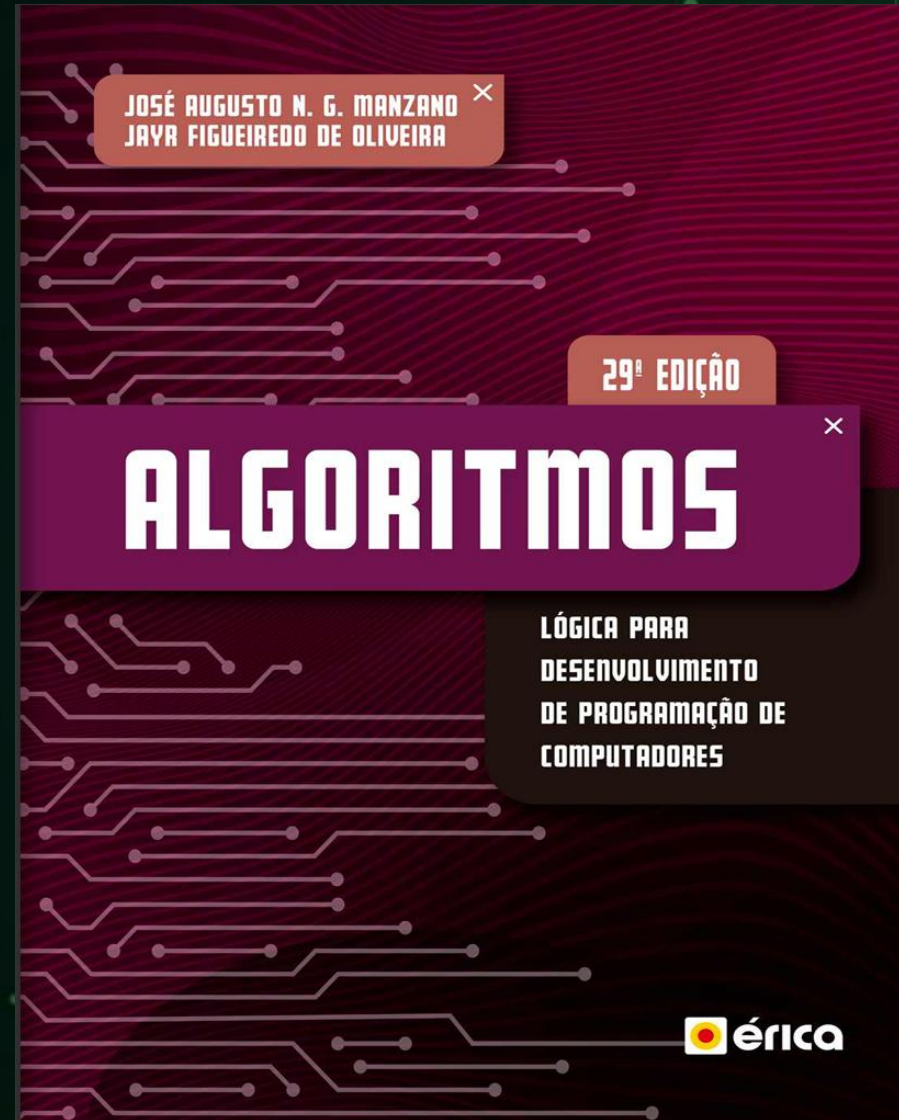
gfsilveira@gmail.com

A alma do software



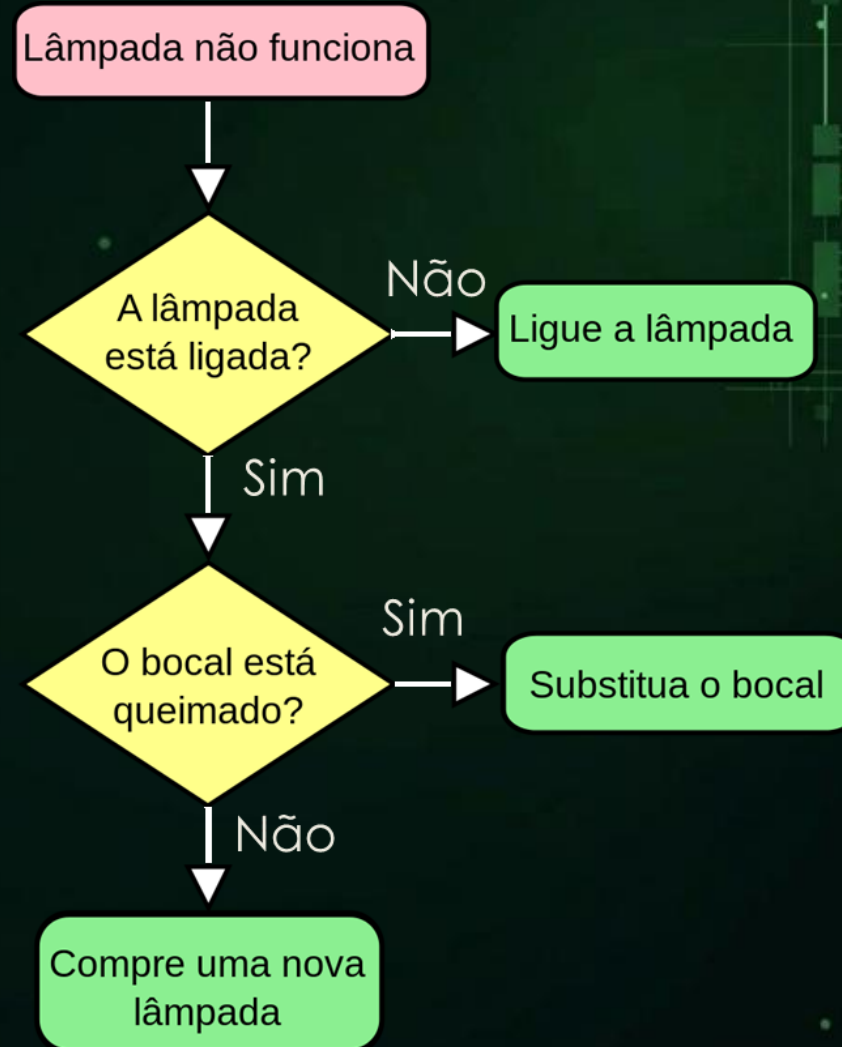
Algoritmo

Algoritmos são conjuntos de passos finitos e organizados que, quando executados, resolvem um determinado problema.



Algoritmo

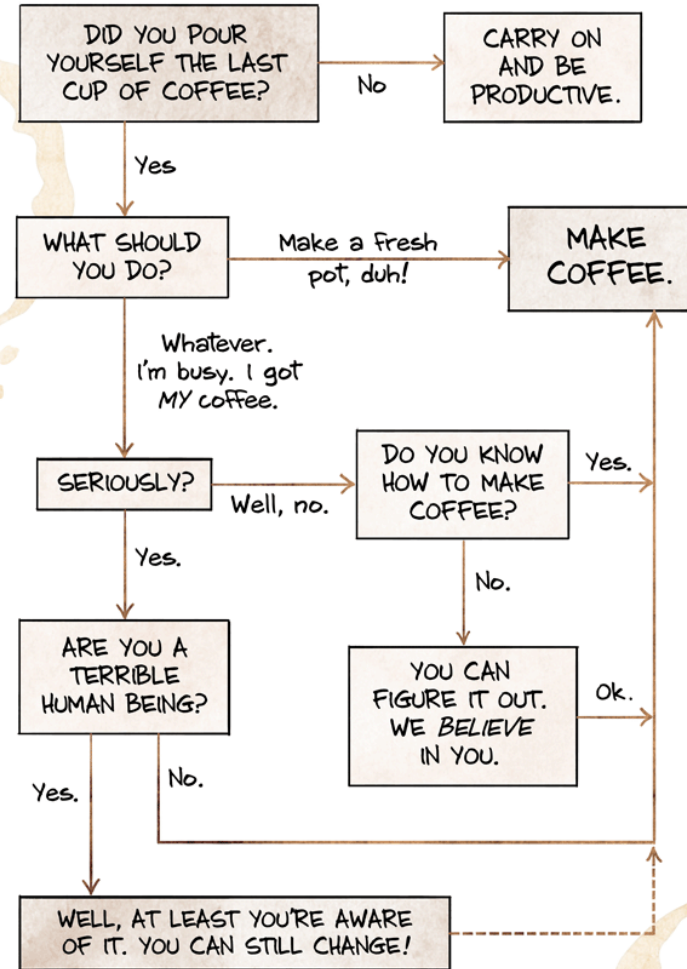
Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.



Algoritmo

Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.

The Office Coffee Flowchart



WWW.PHDCOMICS.COM

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Maior Divisor Comum

Euclides de Alexandria – 400 a.E.C.



$$\text{MDC}(a,b) = \text{MDC}(b,r)$$

Onde:

a = número maior

b = número menor

r = resto de a/b

Como Funciona (Passo a Passo):

- 1.Divisão Inicial:** Divida o maior número pelo menor número.
- 2.Repetição:** Troque o dividendo pelo divisor anterior, e o divisor pelo resto (R). Continue dividindo..
- 3.Finalização:** Repita o processo até que o resto seja 0..
- 4.Resultado:** O MDC é o último resto diferente de zero obtido..

Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.



Sistema numérico hindu-árabe

Abu Abdullah Muhammad bin Musa Al-Khwarizmi – 850

Persian

0, 1, 2, 3, 4, 5, 6, 7, 8, 9



Máquinas de Calcular

Wilhelm Schickard

Arithmeticon Organum - 1623



Blaize Pascal

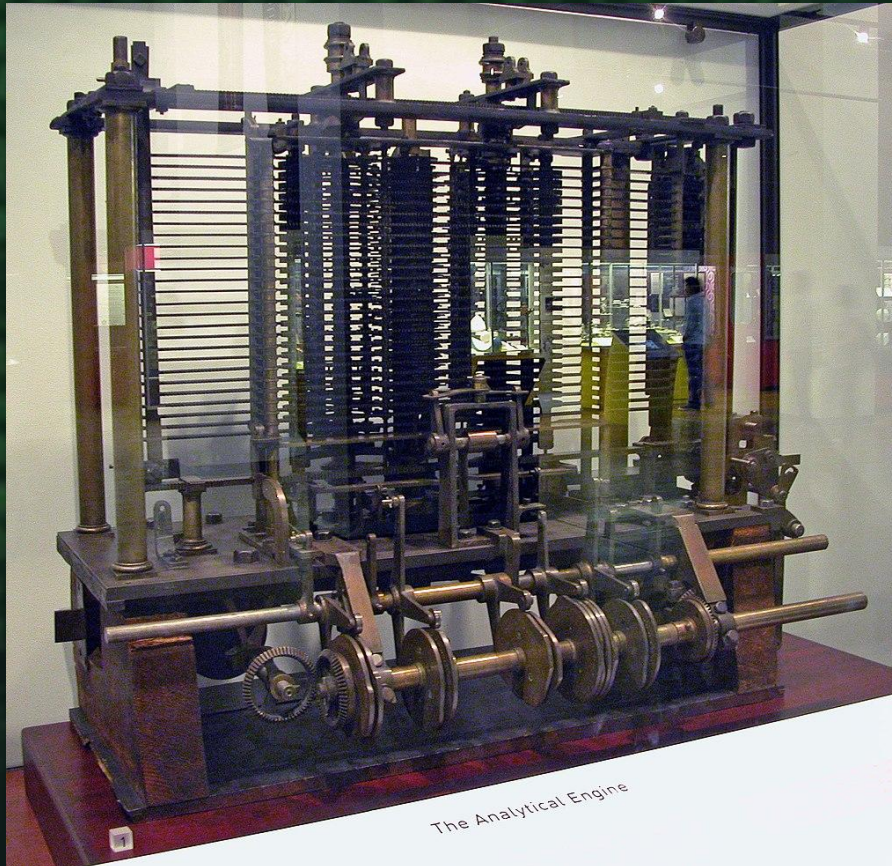
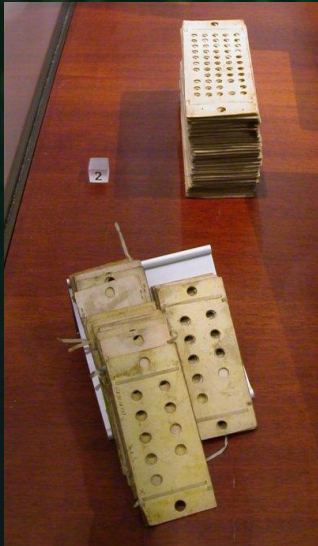
Pascalina - 1642



Motor Analítico

Charles Babbage

“O pai do computador”



1833 – 1910



Primeiro Algoritmo Publicado

Ada L. Lovelace

1840, Babbage, Universidade de Turin -> Luigi Menabrea, Frances

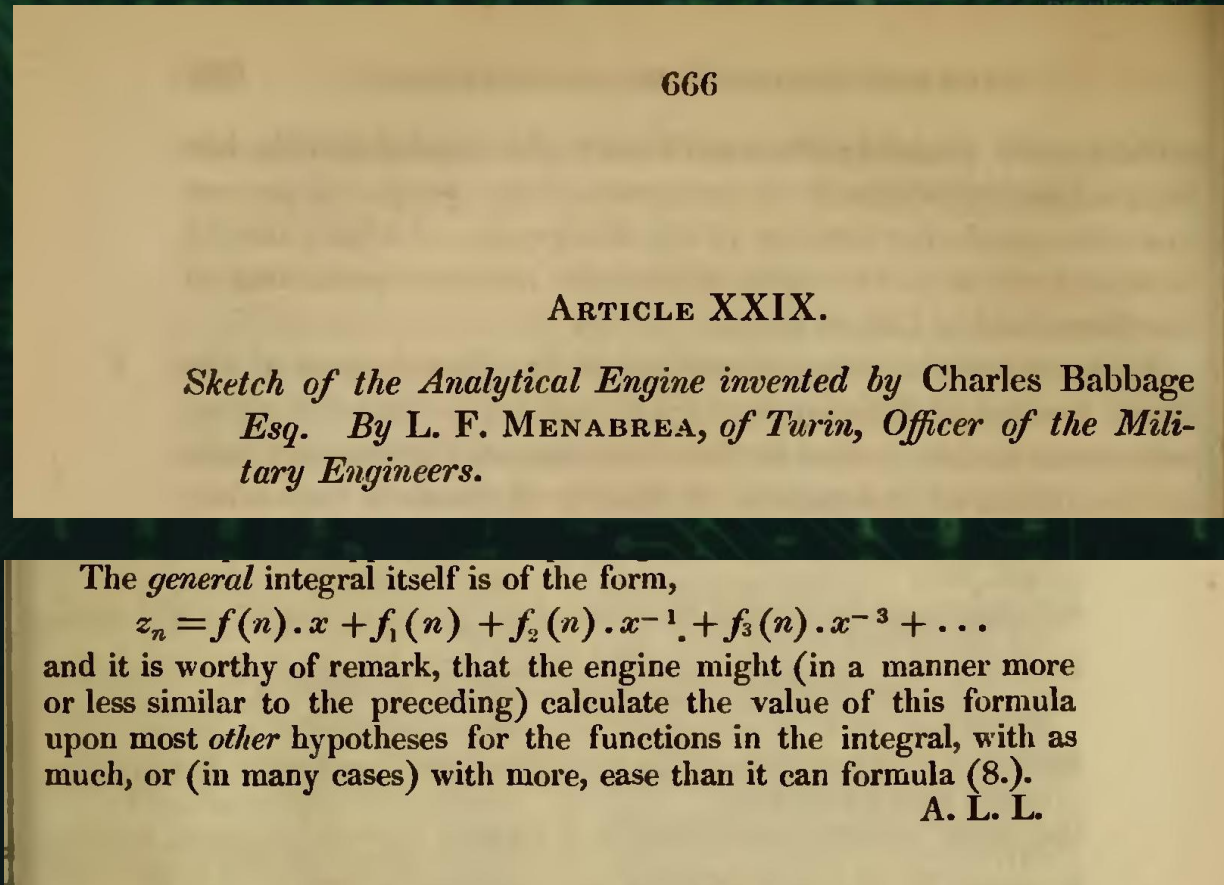
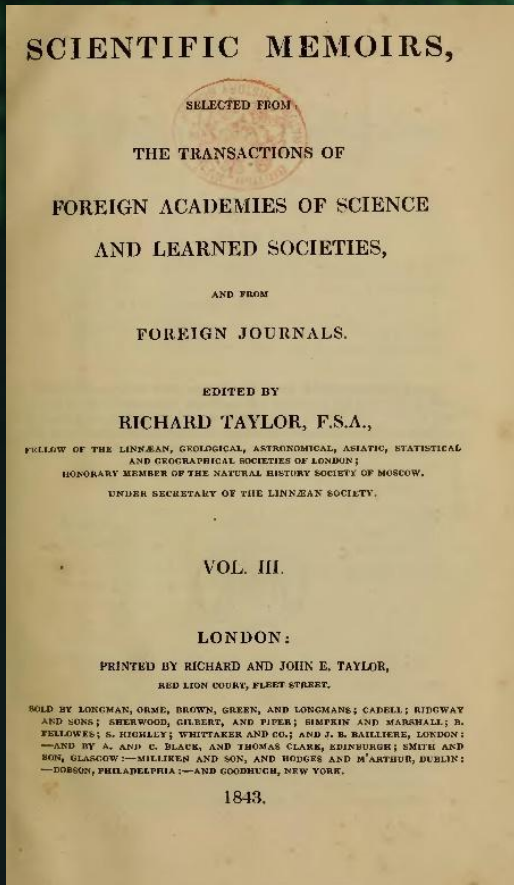
1842, Charles Wheatstone contrata Ada L. Lovelace, Inglês.

1843, publicação.



Primeiro Algoritmo Publicado

Ada L. Lovelace



Primeiro Algoritmo Publicado

Ada L. Lovelace

Nota G, algoritmo para calcular o No de Bernoulli



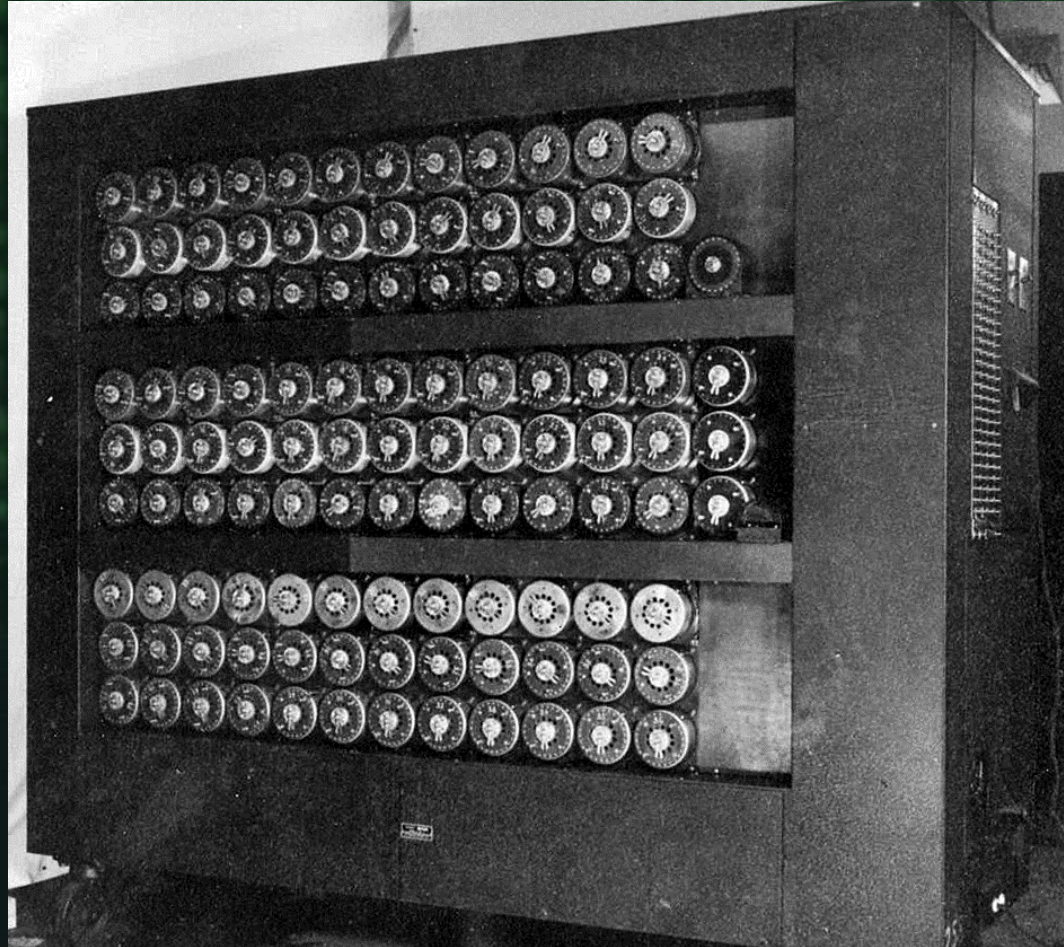
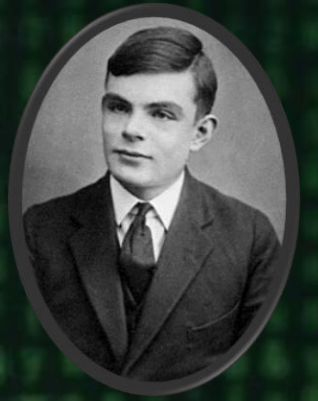
Diagram for the computation by the Engine of the Numbers of Bernoulli. See Note G. (page 722 *et seq.*)

Number of Operation.	Nature of Operation.	Variables acted upon.	Variables receiving results.	Indication of change in the value on any Variable.	Statement of Results.	Data.												Working Variables.										Result Variables.			
						$1V_1$	$1V_2$	$1V_3$	$1V_4$	$1V_5$	$1V_6$	$1V_7$	$1V_8$	$1V_9$	$1V_{10}$	$1V_{11}$	$1V_{12}$	$1V_{13}$	$1V_{14}$	$1V_{15}$	$1V_{16}$	$1V_{17}$	$1V_{18}$	$1V_{19}$	$1V_{20}$	$1V_{21}$	$1V_{22}$	$1V_{23}$	$1V_{24}$		
						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
						1	2	n																							
1	$\times$	$1V_2 \times 1V_3$	$1V_4, 1V_5, 1V_6$	$1V_2 = 1V_2$ $1V_3 = 1V_3$ $1V_4 = 1V_4$ $1V_5 = 1V_5$ $1V_6 = 1V_6$	$= 2n$	...	2	n	2n	2n	2n																				
2	$-$	$1V_4 - 1V_5$	$1V_6$	$1V_6 = 1V_6$	$= 2n - 1$	...	1	...	2n - 1																						
3	$+$	$1V_5 + 1V_6$	$1V_7$	$1V_7 = 1V_7$	$= 2n + 1$	...	1	...	2n + 1																						
4	$+$	$1V_6 + 1V_7$	$1V_8$	$1V_8 = 1V_8$	$= 2n - 1$	...	...	...	0	0																					
5	$+$	$1V_8 + 1V_9$	$1V_{10}$	$1V_{10} = 1V_{10}$	$= 2n + 1$	...	2	...	...	...																					
6	$-$	$1V_{10} - 1V_{11}$	$1V_{12}$	$1V_{12} = 1V_{12}$	$= 2n - 1$	...	...	...	...	...																					
7	$-$	$1V_{12} - 1V_{13}$	$1V_{14}$	$1V_{14} = 1V_{14}$	$= 2n - 1$	...	1	...	n	...																					
8	$+$	$1V_{14} + 1V_{15}$	$1V_{16}$	$1V_{16} = 1V_{16}$	$= 2n + 1$	...	2	...	...	...																					
9	$+$	$1V_{16} + 1V_{17}$	$1V_{18}$	$1V_{18} = 1V_{18}$	$= 2n - 1$	...	...	...	...	...																					
10	$\times$	$1V_{18} \times 1V_{19}$	$1V_{20}$	$1V_{20} = 1V_{20}$	$= 2n + 1$	...	...	...	...	...																					
11	$+$	$1V_{20} + 1V_{21}$	$1V_{22}$	$1V_{22} = 1V_{22}$	$= 2n - 1$	...	...	...	...	...																					
12	$-$	$1V_{22} - 1V_{23}$	$1V_{24}$	$1V_{24} = 1V_{24}$	$= 2n - 1$	...	1	...	...	...																					
13	$+$	$1V_{24} + 1V_{25}$	$1V_{26}$	$1V_{26} = 1V_{26}$	$= 2n + 1$	...	2	...	...	...																					
14	$+$	$1V_{26} + 1V_{27}$	$1V_{28}$	$1V_{28} = 1V_{28}$	$= 2n - 1$	...	...	...	...	...																					
15	$+$	$1V_{28} + 1V_{29}$	$1V_{30}$	$1V_{30} = 1V_{30}$	$= 2n + 1$	...	1	...	...	...																					
16	$\times$	$1V_{30} \times 1V_{31}$	$1V_{32}$	$1V_{32} = 1V_{32}$	$= 2n - 1$	...	...	...	...	...																					
17	$-$	$1V_{32} - 1V_{33}$	$1V_{34}$	$1V_{34} = 1V_{34}$	$= 2n - 1$	...	1	...	...	...																					
18	$+$	$1V_{34} + 1V_{35}$	$1V_{36}$	$1V_{36} = 1V_{36}$	$= 2n + 1$	...	2	...	...	...																					
19	$+$	$1V_{36} + 1V_{37}$	$1V_{38}$	$1V_{38} = 1V_{38}$	$= 2n - 1$	...	...	...	...	...																					
20	$\times$	$1V_{38} \times 1V_{39}$	$1V_{40}$	$1V_{40} = 1V_{40}$	$= 2n + 1$	...	...	...	...	...																					
21	$+$	$1V_{40} + 1V_{41}$	$1V_{42}$	$1V_{42} = 1V_{42}$	$= 2n - 1$	...	...	...	...	...																					
22	$-$	$1V_{42} - 1V_{43}$	$1V_{44}$	$1V_{44} = 1V_{44}$	$= 2n - 1$	...	1	...	...	...																					
23	$-$	$1V_{44} - 1V_{45}$	$1V_{46}$	$1V_{46} = 1V_{46}$	$= 2n - 1$	...	...	...	...	...																					
Here follows a repetition of Operations thirteen to twenty-three.																															
24	$+$	$1V_{46} + 1V_{47}$	$1V_{48}$	$1V_{48} = 1V_{48}$	$= 2n + 1$	...	2	...	...	...																					
25	$+$	$1V_{48} + 1V_{49}$	$1V_{50}$	$1V_{50} = 1V_{50}$	$= 2n - 1$	...	...	...	...	...																					

Máquina de Turing

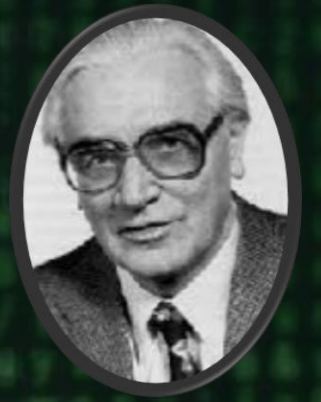
Alan Mathison Turing – 1939-40

The Bombe – 1940

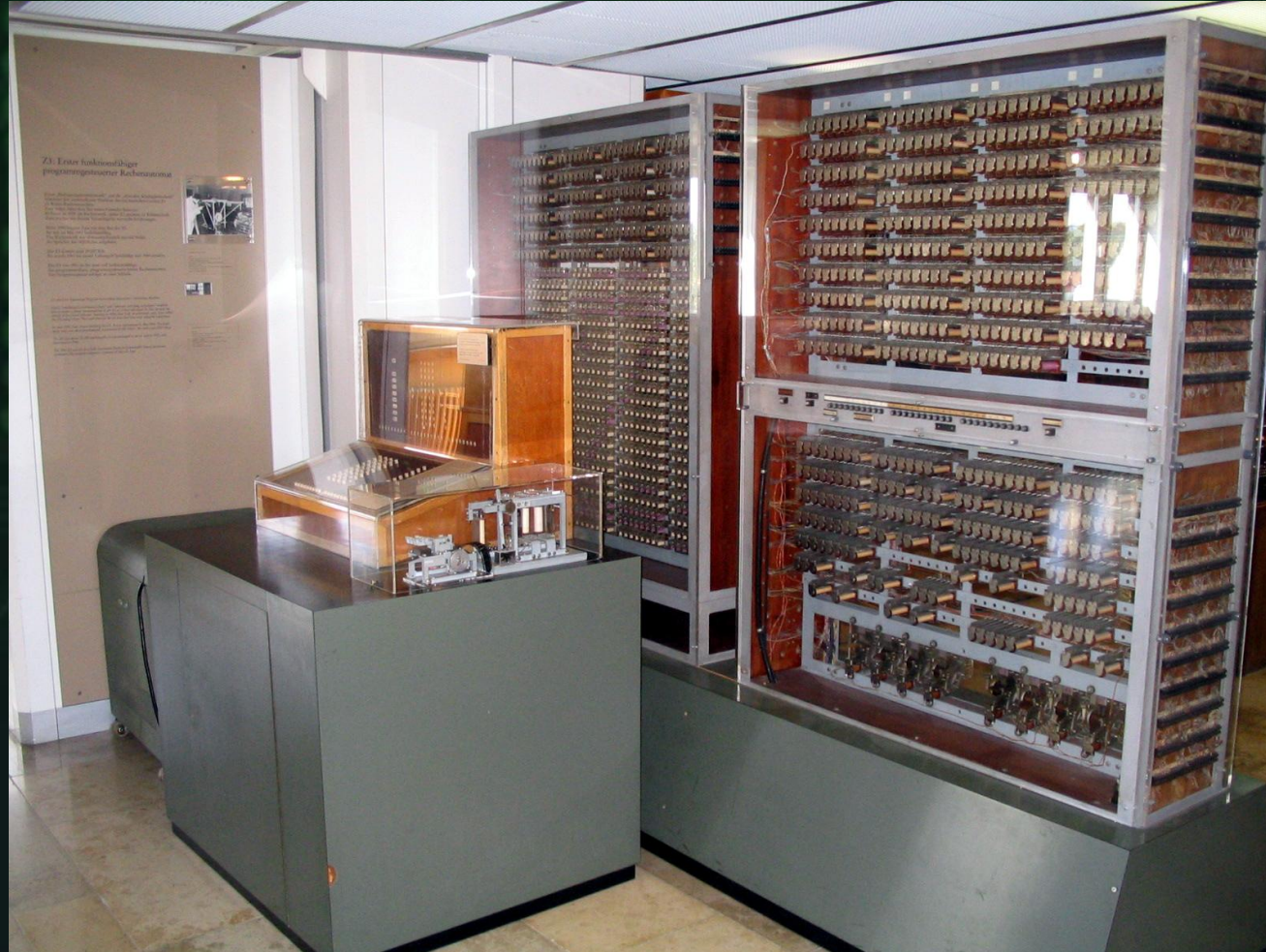


Computador Programável

Konrad Zuse



1941 – Z3



Máquina de Turing

Alan Mathison Turing – 1931



Monatshefte Math. Phys., 38 (1931), 173–198.

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

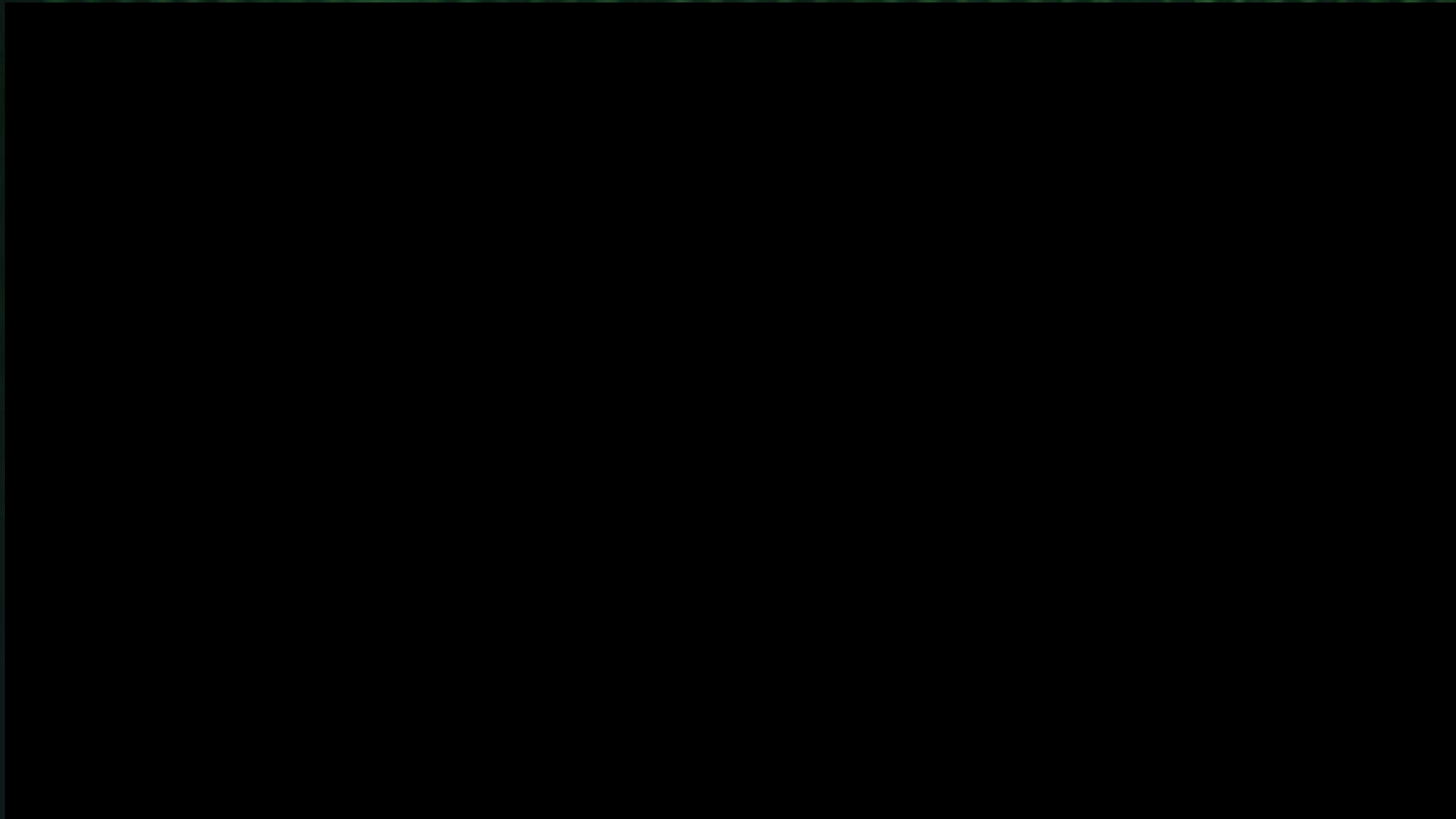
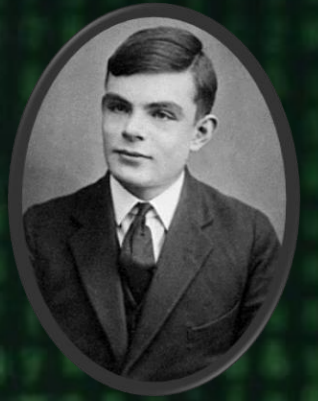
By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]



Máquina de Turing

Alan Mathison Turing – 1931



<http://aturingmachine.com/>, accessed in 12/03/19



Máquina de Turing

Alan Mathison Turing – 1931



<http://aturingmachine.com/>, accessed in 12/03/19



Informação = Bit

Claude Shannon – 1938



A Symbolic Analysis of Relay and Switching Circuits*

*Claude E. Shannon***

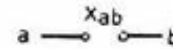


Figure 1 (left). Symbol for hindrance function

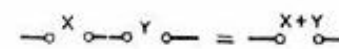


Figure 2 (right). Interpretation of addition

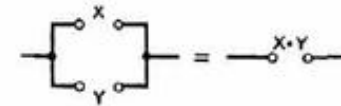


Figure 3 (middle). Interpretation of multiplication

It is evident that with the above definitions the following postulates will hold:

Postulates

1. *a.* $0 \cdot 0 = 0$
b. $1 + 1 = 1$

An open circuit in parallel with a closed circuit is a closed circuit.
An open circuit in series with an open circuit is an open circuit.
2. *a.* $1 + 0 = 0 + 1 = 1$
b. $0 \cdot 1 = 1 \cdot 0 = 0$

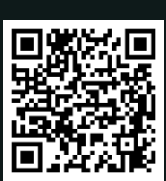
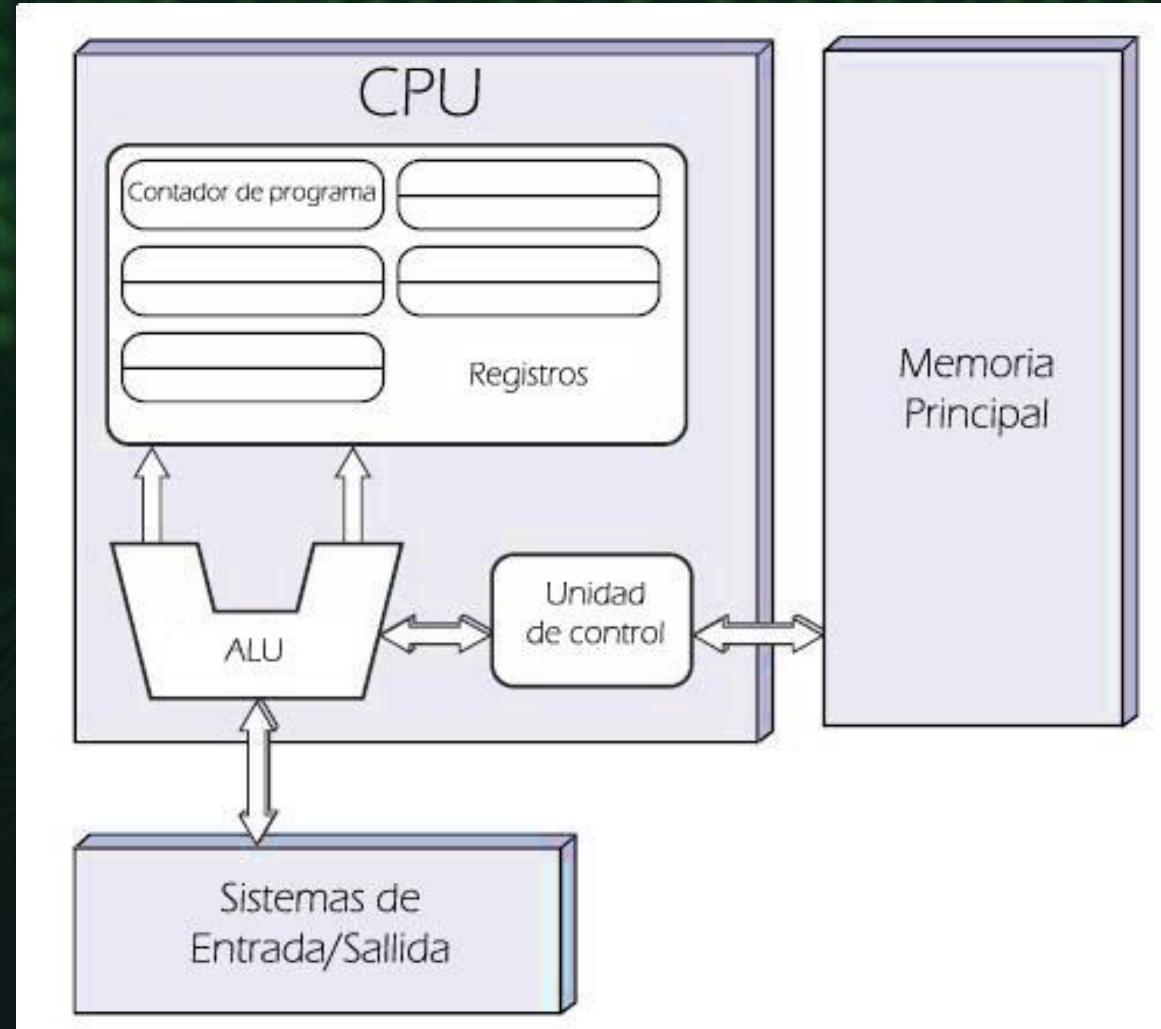
An open circuit in series with a closed circuit in either order (i.e., whether the open circuit is to the right or left of the closed circuit) is an open circuit.
A closed circuit in parallel with an open circuit in either order is a closed circuit.
3. *a.* $0 + 0 = 0$
b. $1 \cdot 1 = 1$

A closed circuit in series with a closed circuit is a closed circuit.
An open circuit in parallel with an open circuit is an open circuit.
4. At any given time either $X = 0$ or $X = 1$.



Infraestructura

John von Neumann – 1945



Programadora da Apollo 11

Margaret Hamilton

1969



Primeiro programa de computador

Contra Alm. Grace M. Hopper – 1954

UNIVAC I (UNIVersal Automatic Computer I)

FLOW-MATIC

```
(0) INPUT INVENTORY FILE-A PRICE FILE-B ; OUTPUT PRICED-INV FILE-C UNPRICED-INV  
FILE-D ; HSP D .  
(1) COMPARE PRODUCT-NO (A) WITH PRODUCT-NO (B) ; IF GREATER GO TO OPERATION 10 ;  
IF EQUAL GO TO OPERATION 5 ; OTHERWISE GO TO OPERATION 2 .  
(2) TRANSFER A TO D .  
(3) WRITE-ITEM D .  
(4) JUMP TO OPERATION 8 .  
(5) TRANSFER A TO C .  
(6) MOVE UNIT-PRICE (B) TO UNIT-PRICE (C) .  
(7) WRITE-ITEM C .  
(8) READ-ITEM A ; IF END OF DATA GO TO OPERATION 14 .  
(9) JUMP TO OPERATION 1 .  
(10) READ-ITEM B ; IF END OF DATA GO TO OPERATION 12 .  
(11) JUMP TO OPERATION 1 .  
(12) SET OPERATION 9 TO GO TO OPERATION 2 .  
(13) JUMP TO OPERATION 2 .  
(14) TEST PRODUCT-NO (B) AGAINST ; IF EQUAL GO TO OPERATION 16 ;  
OTHERWISE GO TO OPERATION 15 .  
(15) REWIND B .  
(16) CLOSE-OUT FILES C ; D .  
(17) STOP . (END)
```



COmmon Business-Oriented Language - COBOL

```
ADD 1 TO x  
ADD 1, a, b TO x ROUNDED, y, z ROUNDED  
  
ADD a, b TO c  
ON SIZE ERROR  
    DISPLAY "Error"  
END-ADD  
  
ADD a TO b  
NOT SIZE ERROR  
    DISPLAY "No error"  
ON SIZE ERROR  
    DISPLAY "Error"
```



Python Básico



5 de abril de 2026